Stefan Zhelyazkov

Stara Zagora, Bulgaria | https://www.stefan-zh.com | stefan.zhelyazkov@gmail.com | https://github.com/stefan-zh/

EDUCATION

University of Pennsylvania, Philadelphia, USA — September 2009 - May 2013
School of Engineering and Applied Science: Bachelor of Science in Engineering (cum laude)
Major: Computer Science
First Minor: Science, Technology & Society — **Second Minor:** Mathematics

WORK EXPERIENCE

Software Architect, Avenga Bulgaria (Musala Soft) — October 2020 – current

- From 2020 to 2025 I delivered software services to SAP Labs Bulgaria:
  - Worked on the Trust Engine project, which is an integration crosspoint between various services. The project fits into a larger software ecosystem and its main goal is to make authentication between services seamless such that technical accounts don't have to be created in each separate service. The project also targets reducing the costs associated with having such technical users in multiple software products. The main technologies used were Golang, Kubernetes and HashiCorp Vault.
  - Worked on adapting the Notary open-source project to work with AWS. The goal was to provide an internal tool which signs and verifies binary artifacts to make sure they haven't been tampered with. The implementation was done in Golang and involved deep understanding of cryptographic principles such as public and private keys and certificates.
  - Developed a plugin for HashiCorp Vault, which signs and verifies JWT tokens. The project was used for internal services.
- My work for Musala Soft included:
  - Helped organize 4 seasons (2021-2025) of the international competitive programming contest CodeIT. Managed the technical team who created the tasks for every season. Alongside that, I was responsible for developing, improving, and maintaining the infrastructure of the competition. Our biggest success was the revamping of the CodeIT Grader (this is the software which scores the submissions), which achieved significant gains: 4.5 times better performance with 60% less code; also, significant improvements in security, stability, CI/CD, extensibility and maintainability. This improved version of the Grader was received well by our contestants.
  - Participated in the recruiting efforts of the company. Conducted more than 60 interviews. Prepped colleagues for client interviews. Did final reviews for candidates; reviewed take-home projects and technical tests.

Software Engineer, WeWork — June 2018 – November 2019

- Worked on a web application for the global furniture inventory of the company. Used Kotlin, Spring, Hibernate, REST, React, TypeScript. Responsibilities included: managing the backend architecture, working with databases and storage, REST APIs, soft delete, security, tests.
- Created a backend library for DAOs (data access objects) which made working with databases and using transactions easier. The library was shared with other teams and received good feedback, which helped its integration in the common backend library for the department.
- Worked with a summer intern whose project was successfully integrated into production.

Software Engineer at Honest Buildings (now part of Procore Technologies) — April 2016 – March 2018

- Developed most of the first Reporting & Exporting tool - DB queries, in-screen data formatting, coloring, filtering, saving, and exporting to Excel. Used Java 8, Dropwizard, AngularJS 1.6, JavaScript, jOOQ.
- Actively contributed to a stronger backend architecture. Initiated and contributed to the separation of the services layer from the endpoints.
- Cleaned up legacy code, resolved bugs, and helped increase code coverage.
- Helped introduce a Java linter and a stylechecker.
- Regularly updated some of our backend library dependencies and performed code uplifting accordingly.

Technology Analyst for the Credit Risk team at Goldman Sachs July 2013 – March 2016

- Responsible for Java back-end applications aggregating GB's of data and producing Credit Risk metrics. Worked on the full flow from sourcing, transforming, aggregating, displaying and storing the data. Optimized the legacy codebase. Documented it. Sped up certain processes by up to 14%.
- Adapted several large Java applications (60GB RAM ~ 120GB) to run on a new grid of servers thus helping reduce maintenance costs. This work resulted in annual savings between $100,000 - $200,000.
- Actively worked with interns and other junior people. Participated in the recruiting efforts of the company, including the firmwide Hackathon Recruiting Committee for 2015/2016.

IN ADDITION

- Completed Coursera course Building Cloud Services with Java and Spring, October 2019
- Completed Coursera course Kotlin for Java Developers, by JetBrains, December 2018
- Completed Coursera course Front-End Web Development with React, November 2018
- Completed Coursera course on Algorithms, part I, by Stanford University, Fall 2014
- Completed Coursera course on Algorithms, part II, by Stanford University, Spring 2015
- Completed Coursera course on Functional Programming Principles in Scala, Spring 2014
- Solved 95 problems on Project Euler (Level 3). Top 1.379% of more than 1MM registered users with at least 1 solved problem.

PROGRAMMING LANGUAGES

Java (10+ years), Golang (5+ years), JavaScript/TypeScript (5+ yrs), Kotlin (3+ yrs), Python, PHP, Scala, Haskell, OCaml and others.
Platforms and libraries: Spring, React, Dropwizard, Hibernate, jOOQ, Junit
Clouds: AWS, GCP